OXFORD AQA

INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number | | | | | |　　Candidate number | | | | |

Surname ______________________________

Forename(s) ______________________________

Candidate signature ______________________________

I declare this is my own work.

INTERNATIONAL A-LEVEL
CHEMISTRY (9620)

Unit 5: Practical and synoptic

Monday 16 June 2025　　07:00 GMT　　Time allowed: 1 hour 25 minutes

Materials

For this paper you must have:
- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- The marks for questions are shown in brackets.
- The maximum mark for this paper is 60.

For Examiner's Use	
Question	Mark
1	
2	
3	
4–33	
TOTAL	

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There are no questions printed on this page

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ANSWER IN THE SPACES PROVIDED

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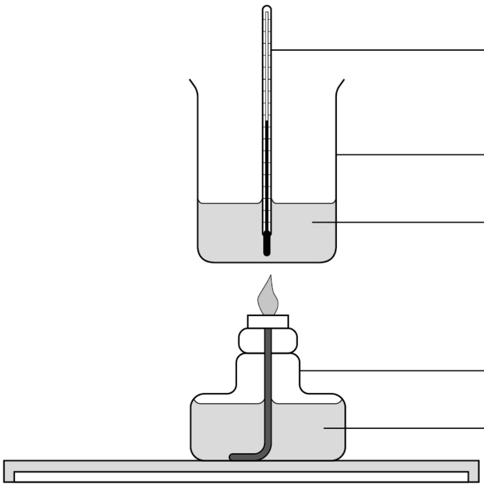
Section A

Answer **all** questions in the spaces provided.

0 1 This question is about combustion of alcohols.

Figure 1 shows the equipment a student uses to determine the enthalpy of combustion ($\Delta_c H$) of butan-1-ol.

Figure 1



0 1 . 1 Suggest **one** improvement to the equipment in **Figure 1** to minimise heat loss from the experiment.

[1 mark]

____

____

Question 1 continues on the next page

Table 1 shows the student's data from their experiment.

Table 1

Volume of water	200 cm^3
Initial temperature	17.2 °C
Final temperature	43.7 °C
Initial mass of spirit burner and butan-1-ol	59.40 g
Final mass of spirit burner and butan-1-ol	58.60 g

0 1 . 2 Use the data in **Table 1** to calculate $\Delta_c H$, in kJ mol^{-1}, of butan-1-ol.

Density of water = 1.00 g cm^{-3}
Specific heat capacity of water = 4.18 J K^{-1} g^{-1}

[3 marks]

$\Delta_c H$ ________________________ kJ mol^{-1}

Do not write outside the box

The student repeats the experiment using the same mass of butan-1-ol but a larger volume of water.

0 1 . 3 Which is the correct change in the percentage uncertainty of the measurement of the temperature rise?

Tick (✓) **one** box.

[1 mark]

Increases	**Stays the same**	**Decreases**

0 1 . 4 Suggest why using the same mass of butan-1-ol but a larger volume of water reduces the heat loss from the calorimeter and water.

[1 mark]

____

____

0 1 . 5 The student observed that a black solid appeared on the bottom of the calorimeter.

Identify the black solid.
Explain why it is formed.

[2 marks]

Black solid ____

Explanation ____

____

Question 1 continues on the next page

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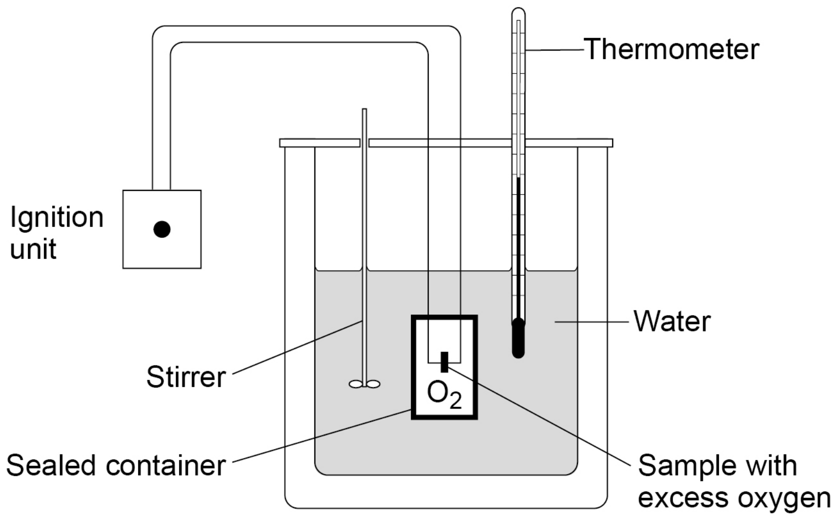
0 1 . 6 **Figure 2** shows different equipment.

Figure 2



Suggest why the equipment in **Figure 2** could not be used to measure an enthalpy change.
Do **not** answer in terms of safety.

[1 mark]

____

____

Table 2 shows some bond enthalpy values.

Table 2

Type of bond	C–C	C–H	C–O	C=O	O–H	O=O
Bond enthalpy / kJ mol^{-1}	348	412	360	805	463	496

Table 3 shows some enthalpy of vaporisation values.

Table 3

Substance	**Enthalpy of vaporisation / kJ mol^{-1}**
Propan-1-ol	41.4
Water	40.7

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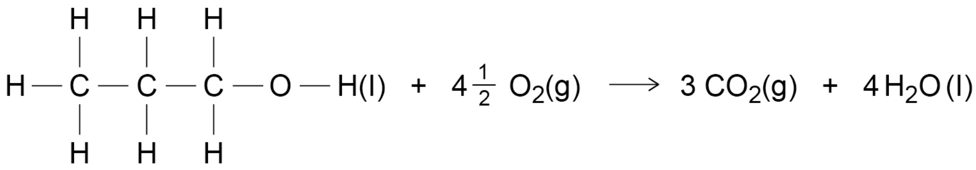
0 1 . 7 Use the data in **Table 2** and **Table 3** to calculate a value for $\Delta_c H$, in kJ mol^{-1}, for liquid propan-1-ol.



Enthalpy of vaporisation of propan-1-ol = 41.4 kJ mol^{-1}
Enthalpy of vaporisation of water = 40.7 kJ mol^{-1}

[3 marks]

$\Delta_c H$ ______________________ kJ mol^{-1}

0 1 . 8 Suggest **one** reason why your answer to Question **01.7** is different from the data book value for $\Delta_c H$ of liquid propan-1-ol.

[1 mark]

____

____

13

Turn over ►

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0 2 A student sets up a $Cu^{2+}(aq) / Cu(s)$ electrode by placing a piece of copper metal in a beaker containing copper(II) sulfate solution.

The student measures the standard electrode potential of the $Cu^{2+}(aq) / Cu(s)$ electrode by connecting it to a standard hydrogen electrode containing $H_2SO_4(aq)$

0 2 . 1 State the conditions used in the standard hydrogen electrode.

[2 marks]

Concentration of $H_2SO_4(aq)$ ____________________

Temperature of $H_2SO_4(aq)$ ____________________

Pressure of $H_2(g)$ ____________________

0 2 . 2 Give **one** reason why platinum is used as the electrode in the standard hydrogen electrode.

[1 mark]

____

____

0 2 . 3 Give the conventional representation of the cell used to measure the standard electrode potential of the $Cu^{2+}(aq) / Cu(s)$ electrode.

[1 mark]

____

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The standard electrode potential ($E^\ominus$) for the $Cu^{2+}(aq)/Cu(s)$ electrode is +0.34 V

The $Cu^{2+}(aq)/Cu(s)$ electrode is connected to other standard electrodes made of metals in contact with their ions in solution.

Table 4 shows the EMFs of the cells.

Table 4

Negative electrode	Positive electrode	EMF / V
$Zn^{2+}(aq)/Zn(s)$	$Cu^{2+}(aq)/Cu(s)$	+ 1.10
$Ni^{2+}(aq)/Ni(s)$	$Cu^{2+}(aq)/Cu(s)$	+ 0.59
$Pb^{2+}(aq)/Pb(s)$	$Cu^{2+}(aq)/Cu(s)$	+ 0.47

0 2 . 4 Identify the species in **Table 4** that is the most powerful reducing agent.

[1 mark]

____

0 2 . 5 Calculate $E^\ominus$ of the $Ni^{2+}(aq)/Ni(s)$ electrode.

[1 mark]

______________________________ V

Question 2 continues on the next page

Table 4 is repeated here.

Table 4

Negative electrode	Positive electrode	EMF / V
$Zn^{2+}(aq) / Zn(s)$	$Cu^{2+}(aq) / Cu(s)$	+ 1.10
$Ni^{2+}(aq) / Ni(s)$	$Cu^{2+}(aq) / Cu(s)$	+ 0.59
$Pb^{2+}(aq) / Pb(s)$	$Cu^{2+}(aq) / Cu(s)$	+ 0.47

An electrochemical cell is made by connecting the $Zn^{2+}(aq) / Zn(s)$ and $Pb^{2+}(aq) / Pb(s)$ electrodes.

0 2 . 6 Use the data in **Table 4** to calculate the EMF of this cell.

[1 mark]

____ V

0 2 . 7 A salt bridge is used to connect the electrodes.

State the purpose of the salt bridge.

Suggest a substance that could be used in the salt bridge of this cell.

[2 marks]

Purpose of the salt bridge ____

____

____

Substance in the salt bridge ____

____

9

0 3 This question is about the preparation of pure cinnamic acid ($C_6H_5CH{=}CHCOOH$) which is a carboxylic acid.

Cinnamic acid is a solid at room temperature and is insoluble in cold water.
It is prepared in several stages:

Stage 1
Benzaldehyde is heated and reacts with hot ethanoic anhydride.
After cooling, an excess of aqueous sodium carbonate solution is added to the product.

A solution containing sodium cinnamate ($C_6H_5CH{=}CHCOONa$) and sodium ethanoate is formed.

Stage 2
Concentrated hydrochloric acid is added slowly to this solution.
Cinnamic acid is formed.

$$C_6H_5CH{=}CHCOONa(aq) + HCl(aq) \rightarrow C_6H_5CH{=}CHCOOH(s) + NaCl(aq)$$

Stage 3
Solid cinnamic acid is collected by filtration.
The solid is washed with water.

0 3 . 1 The heating in **Stage 1** is done with a hot water bath.

Suggest why a Bunsen burner is **not** used.

[1 mark]

____

____

0 3 . 2 Suggest why, in **Stage 2**, it is important to add the concentrated hydrochloric acid slowly to the solution from **Stage 1**.

[1 mark]

____

____

Question 3 continues on the next page

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0 3 . 3 In **Stage 2**, the sodium ethanoate in the mixture is also converted into a carboxylic acid.
During **Stage 3** this carboxylic acid is removed.

Suggest why this carboxylic acid is removed even though the solid cinnamic acid remains on the filter paper.

[1 mark]

____

____

Stage 4
The impure cinnamic acid is purified by recrystallisation from ethanol.

Stage 5
Crystals of pure cinnamic acid are collected by filtration under reduced pressure.

0 3 . 4 Suggest why ethanol is a suitable solvent for the recrystallisation in **Stage 4**.

[1 mark]

____

____

0 3 . 5 Give **two** advantages of filtration under reduced pressure in **Stage 5**.

[2 marks]

Advantage 1 ____

____

Advantage 2 ____

____

Do not write outside the box

0 3 . 6 The student measures the melting point of their sample of cinnamic acid and finds that it melts over a temperature range between 125 °C and 132 °C
The data book value for cinnamic acid is 133 °C

Suggest why the student's sample melted over a range.

[1 mark]

0 3 . 7 Use IUPAC rules to name cinnamic acid ($C_6H_5CH{=}CHCOOH$)

[1 mark]

8

Turn over for the next section

Do not write outside the box

Section B

Each question is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.
For each question completely fill in the circle alongside the appropriate answer.

CORRECT METHOD   WRONG METHODS

If you want to change your answer you must cross out your original answer as shown.

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.

You may do your working in the blank space around each question but this will not be marked.

0 4 All of the molecules shown have polar bonds.

Which molecule does **not** have a permanent dipole?

[1 mark]

A CH_2Cl_2 ○

B SbH_3 ○

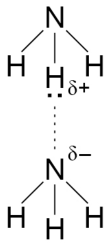
C $CHCl_3$ ○

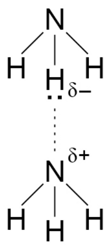
D $AsCl_5$ ○

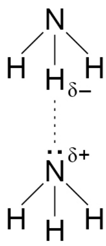
Do not write outside the box

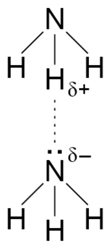
0 5 Which of these correctly shows a hydrogen bond between two molecules of ammonia?

[1 mark]

A 

B 

C 

D 

Turn over ►

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0 6 In which equation is nitrogen reduced?

[1 mark]

A $2Na_3N \rightarrow 6Na + N_2$ ⬭

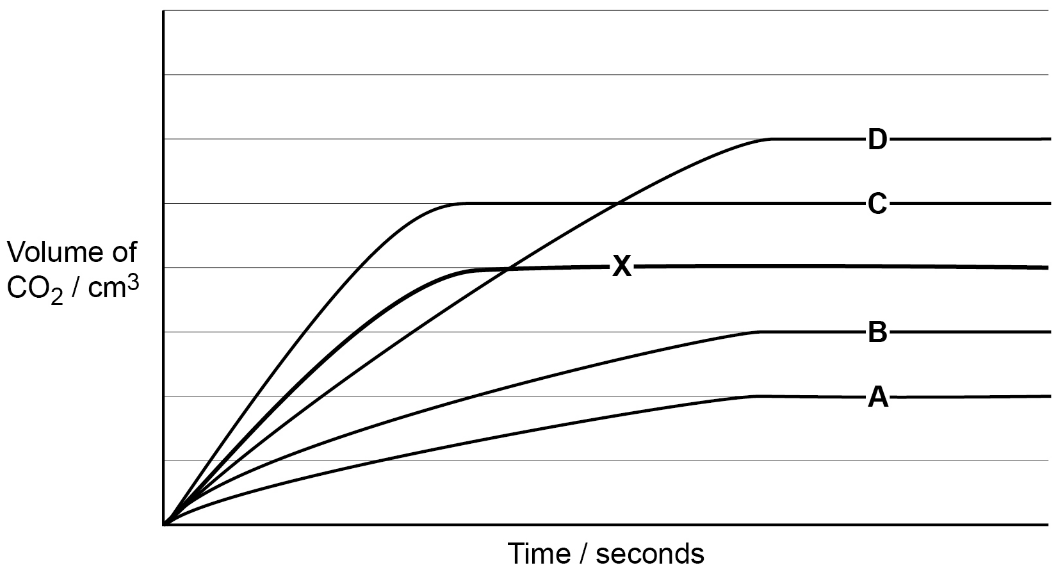
B $2NO + 2CO \rightarrow N_2 + 2CO_2$ ⬭

C $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ ⬭

D $HNO_3 + H_2SO_4 \rightarrow NO_2^+ + H_2O + HSO_4^-$ ⬭

0 7 The reaction between calcium carbonate and hydrochloric acid produces CO_2 gas.

Line X, on the graph below, shows how the volume of CO_2 changes with time as excess calcium carbonate reacts with 10 cm^3 of 2.0 mol dm^{-3} hydrochloric acid.



The reaction is then repeated with excess calcium carbonate and 15 cm^3 of 1.0 mol dm^{-3} hydrochloric acid. All other conditions remain the same.

Which line shows how the volume of CO_2 changes over time?

[1 mark]

A Line A ⬭

B Line B ⬭

C Line C ⬭

D Line D ⬭

Do not write outside the box

0 8 A gaseous reaction mixture reaches equilibrium in a sealed container at a given temperature.

The experiment is repeated under the same conditions but in a smaller container.

Which is correct about the second experiment?

[1 mark]

A A larger percentage of the particles has an energy greater than the activation energy. ⬭

B The kinetic energy of all particles increases, leading to more collisions. ⬭

C The percentage of particles with the mean energy remains the same. ⬭

D The percentage of particles with the most probable energy increases. ⬭

0 9 Which statement is **not** correct about the role of catalysts in equilibrium reactions?

[1 mark]

A Catalysts affect the value of the equilibrium constant K_c ⬭

B Catalysts increase the rate of the forward reaction and the rate of the reverse reaction. ⬭

C Catalysts provide an alternative pathway with a lower E_a ⬭

D Catalysts take part in the reaction but are not used up. ⬭

Turn over for the next question

Do not write outside the box

1 0 Which is the expression for the equilibrium constant (K_c) for this equilibrium?

$$2\,CHClF_2(g) \rightleftharpoons C_2F_4(g) + 2\,HCl(g)$$

[1 mark]

A $K_c = \dfrac{[CHClF_2]^2}{[C_2F_4] \times [HCl]^2}$ ⬭

B $K_c = \dfrac{[CHClF_2]}{[C_2F_4] \times [HCl]}$ ⬭

C $K_c = \dfrac{[C_2F_4] + [HCl]^2}{[CHClF_2]^2}$ ⬭

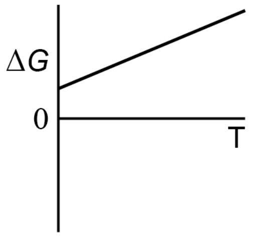
D $K_c = \dfrac{[C_2F_4] \times [HCl]^2}{[CHClF_2]^2}$ ⬭

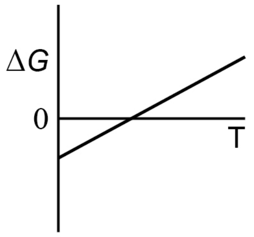
Do not write outside the box

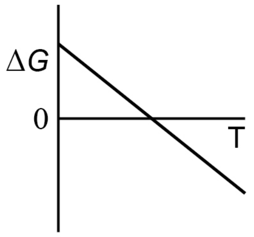
1 1 Which shows how the Gibbs free-energy change (ΔG) varies with temperature (T) for the reaction of magnesium with oxygen?

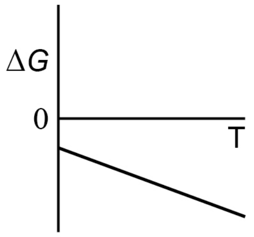
$$2\,Mg(s) + O_2(g) \rightarrow 2\,MgO(s)$$

[1 mark]

A 

B 

C 

D 

Turn over ►

1 2 Which solution has the lowest pH?

[1 mark]

A A solution of 0.00154 mol dm^{-3} hydrochloric acid

B An equal mixture of 0.00154 mol dm^{-3} hydrochloric acid and water

C A solution of 0.146 mol dm^{-3} butanoic acid (K_a of butanoic acid = 1.51×10^{-5} mol dm^{-3})

D A solution with pH = 2.84

1 3 Which is true about the rate constant, *k*?

[1 mark]

A *k* decreases when the total pressure increases because the collision frequency increases.

B *k* decreases when the temperature increases because the collision frequency increases.

C *k* increases when the temperature increases because more particles have $E \geq E_a$

D *k* remains unchanged when the temperature increases and the pressure decreases.

1 4 Which statement is true about this equilibrium?

$$X(g) + Y(g) \rightleftharpoons Z(g) \qquad \Delta H = +200 \text{ kJ mol}^{-1}$$

[1 mark]

A The amount of Z remains constant when pressure increases.

B K_p decreases when pressure increases.

C K_p increases when temperature increases.

D The amount of Z remains constant when temperature increases.

Do not write outside the box

1 5 Which statement about the elements of Period 3 is correct?

[1 mark]

A Magnesium has a higher melting point than sodium because magnesium atoms are larger than sodium atoms. ⬭

B Silicon has the highest melting point because it has very strong van der Waals forces between its molecules. ⬭

C Sulfur has a lower melting point than chlorine because it has more covalent bonds in a molecule. ⬭

D Argon has the lowest melting point because it only has van der Waals forces between its atoms. ⬭

1 6 A teacher adds some Group 2 metals to an excess of dilute sulfuric acid in two separate beakers, **P** and **Q**.

The same volume and concentration of sulfuric acid is in each beaker.

0.15 g of fresh magnesium powder is added to beaker **P**.

0.15 g of fresh barium powder is added to beaker **Q**.

Some measurements and observations are made.

Which is correct?

[1 mark]

A The mass change of beaker **P** and its contents is greater than the mass change of beaker **Q** and its contents. ⬭

B The final pH is lower in beaker **P** than the final pH in beaker **Q**. ⬭

C The reaction in beaker **P** starts more violently than the reaction in beaker **Q**. ⬭

D A white precipitate is seen in beaker **P** but not in beaker **Q**. ⬭

Do not write outside the box

1 7 A student reacts propanoic anhydride with water.

Which is most suitable to dry the product?

[1 mark]

A Magnesium oxide ⬭

B Calcium sulfate ⬭

C Strontium carbonate ⬭

D Barium hydroxide ⬭

1 8 Fluorine is the element with the highest electronegativity value.

Which statement about electronegativity is correct?

[1 mark]

A Electronegativity decreases down a group because there are more electron shells and more shielding in the atoms. ⬭

B Electronegativity increases across a period because there is a bigger nuclear charge and the atoms get bigger. ⬭

C Electronegativity is the power of an atom to attract an electron towards itself. ⬭

D Electronegativity of an atom is the energy released when an electron is added to a gaseous atom. ⬭

1 9 Magnesium chloride is added to a beaker of cold water.

Which statement is correct?

[1 mark]

A A colourless alkaline solution is formed. ⬭

B Hydrogen bonds form between $MgCl_2$ and polar water molecules. ⬭

C The solution formed conducts electricity. ⬭

D White clouds of magnesium oxide are observed. ⬭

Do not write outside the box

2 0 Which shows the oxides of sodium, aluminium, silicon and sulfur in order of **increasing** melting point?

[1 mark]

A $Na_2O < SiO_2 < SO_3 < Al_2O_3$ ○

B $SiO_2 < Na_2O < Al_2O_3 < SO_3$ ○

C $SO_3 < Al_2O_3 < Na_2O < SiO_2$ ○

D $SO_3 < Na_2O < Al_2O_3 < SiO_2$ ○

2 1 A student obtains the highest oxides of some Period 3 elements.
The student then adds water to each oxide and measures the pH of the mixture formed.

Which element forms a white solid that can give a colourless solution of pH = 13?

[1 mark]

A Na ○

B Al ○

C Si ○

D S ○

Turn over for the next question

Do not write outside the box

2 2 Which row correctly identifies the type of isomerism shown by the complex ion?

[1 mark]

		cis-trans **isomerism**	**optical isomerism**	
A	$[Co(NH_2CH_2CH_2NH_2)_3]^{2+}$	No	No	⬭
B	$[Cu(NH_3)_4(H_2O)_2]^{2+}$	No	Yes	⬭
C	$FeCl_4^-$	Yes	No	⬭
D	$[Fe(C_2O_4)_2(H_2O)_2]^{2-}$	Yes	Yes	⬭

2 3 A 25.0 cm^3 sample of 0.0500 mol dm^{-3} sodium ethanedioate solution ($Na_2C_2O_4$) needs 38.50 cm^3 of acidified potassium dichromate(VI) solution for a complete reaction.

$$Cr_2O_7^{2-} + 3C_2O_4^{2-} + 14H^+ \rightarrow 2Cr^{3+} + 7H_2O + 6CO_2$$

Which is the concentration of the potassium dichromate(VI) solution?

[1 mark]

A 0.0108 mol dm^{-3} ⬭

B 0.0257 mol dm^{-3} ⬭

C 0.0325 mol dm^{-3} ⬭

D 0.0974 mol dm^{-3} ⬭

Do not write outside the box

2 4 Chromium can exist as both Cr^{2+} and Cr^{3+} ions.
These behave as typical M^{2+} and M^{3+} ions in aqueous solution.

Which is correct about 0.1 mol dm^{-3} aqueous solutions of the metal aqua ions $[Cr(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{3+}$?

[1 mark]

A $[Cr(H_2O)_6]^{2+}$ is more acidic because Cr^{2+} has a smaller ionic radius than Cr^{3+} ⬭

B $[Cr(H_2O)_6]^{3+}$ is more acidic because Cr^{3+} has a higher charge:size ratio than Cr^{2+} ⬭

C $[Cr(H_2O)_6]^{2+}$ and $[Cr(H_2O)_6]^{3+}$ each react with aqueous sodium carbonate to form carbon dioxide gas ⬭

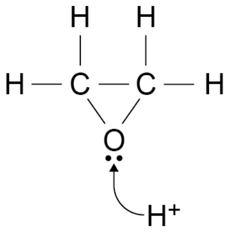
D The pH of aqueous $[Cr(H_2O)_6]^{3+}$ is higher than that of $[Cr(H_2O)_6]^{2+}$ ⬭

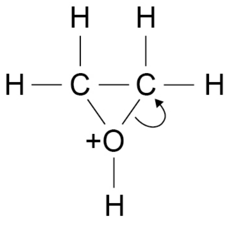
Turn over for the next question

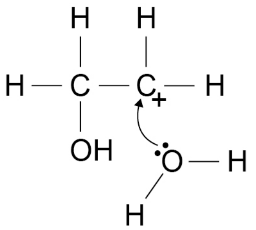
2 5 Epoxyethane reacts with water, in the presence of acid, to produce ethane-1,2-diol.

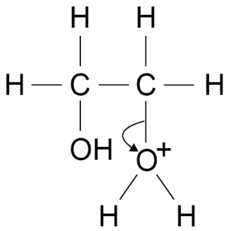
Which is a correct step in the mechanism?

[1 mark]

A  ⬭

B  ⬭

C  ⬭

D  ⬭

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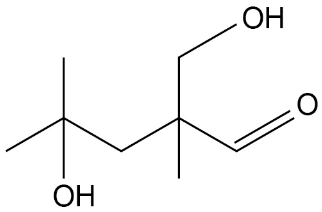
2 6 Compound **J** is heated under reflux with an excess of acidified potassium dichromate(VI) and is oxidised to compound **T**.

Compound J



The equation for this oxidation of **J** can be written as

$$\mathbf{J} + m[O] \rightarrow \mathbf{T} + n H_2O$$

[O] represents the oxidising agent in the equation.

Which shows the correct values for m and n in the balanced equation?

[1 mark]

A m = 4 and n = 2

B m = 3 and n = 1

C m = 2 and n = 2

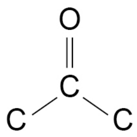
D m = 1 and n = 1

2 7 Which is correct about the reaction between butanone and KCN followed by dilute acid?

[1 mark]

A The first step in the mechanism is electrophilic attack by a cyanide ion.

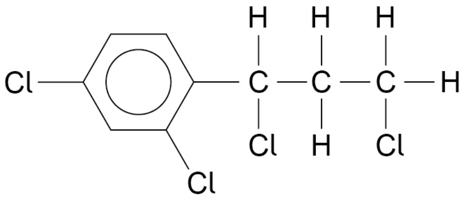
B The mechanism is addition–elimination.

C The product is a single enantiomer because the  group is planar.

D The product is 2-hydroxy-2-methylbutanenitrile.

Do not write outside the box

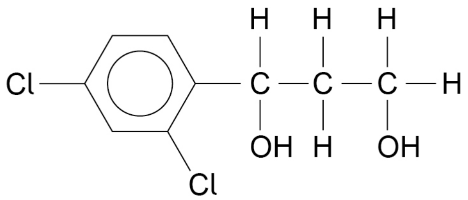
2 8 1 mol of compound **Z** is warmed with 2 mol of aqueous sodium hydroxide.

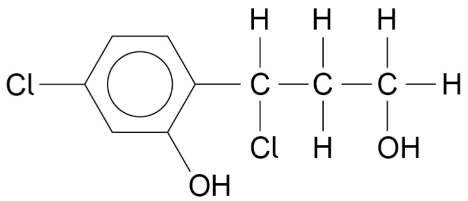


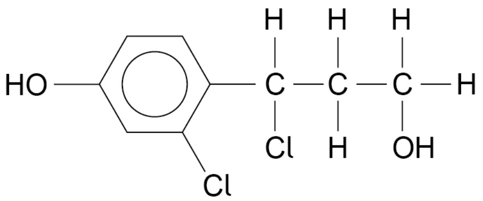
Z

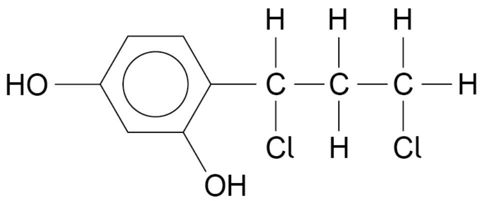
Which compound is formed in the greatest amount?

[1 mark]

A  ⬭

B  ⬭

C  ⬭

D  ⬭

2 9 0.1 mol dm^{-3} solutions of the following amines are prepared.

Which solution has the lowest pH?

[1 mark]

A CH_3NH_2 ⬭

B $C_6H_5NH_2$ ⬭

C $C_6H_5NHCH_3$ ⬭

D $C_6H_5CH_2NH_2$ ⬭

Turn over for the next question

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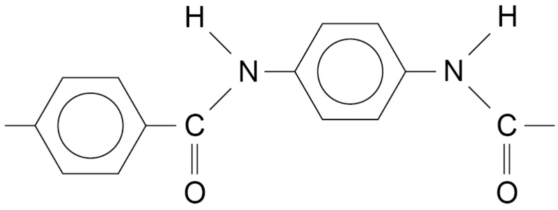
3 0 Kevlar is a polymer made from benzene-1,4-diamine and benzene-1,4-dicarboxylic acid.

Which diagram shows one repeating unit of Kevlar?

[1 mark]

A



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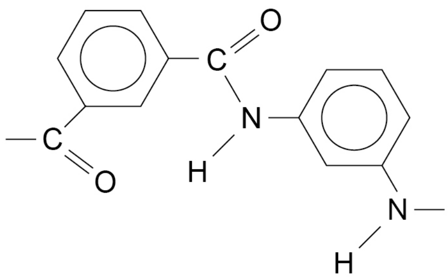

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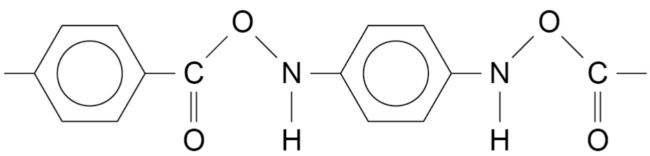
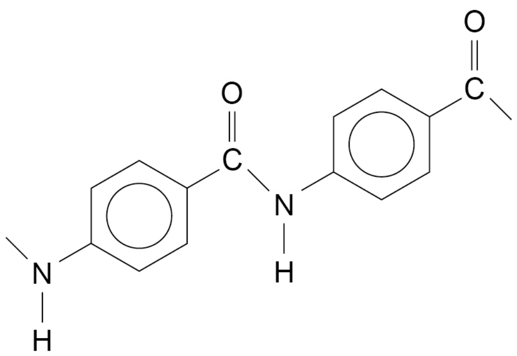


D



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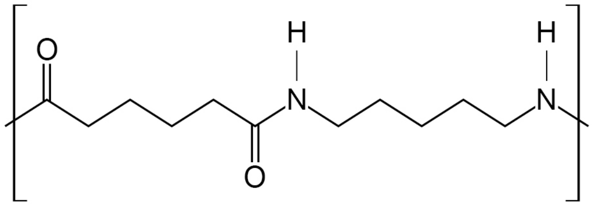
3 1 The repeating unit of a polymer is shown.



Which pair of molecules could be used to make this polymer?

[1 mark]

A Pentane-1,1-diamine and hexanedioic acid ⬭

B Pentane-1,5-diamine and hexanedioic acid ⬭

C Hexane-1,6-diamine and pentanedioic acid ⬭

D 5-Aminopentanoic acid and 6-aminohexanoic acid ⬭

3 2 Which describes how polyesters are hydrolysed by aqueous sodium hydroxide?

[1 mark]

A Electrophilic attack on the oxygen of the carbonyl group ⬭

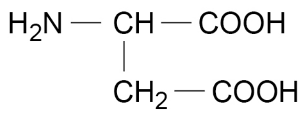
B Nucleophilic attack on the carbon of the carbonyl group ⬭

C Oxidation of the carbonyl group to form carboxylic acids ⬭

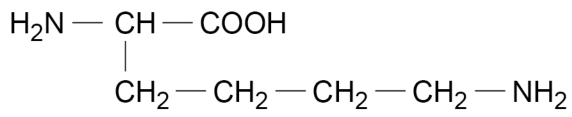
D Reduction of the carbonyl group to form alcohols ⬭

Turn over for the next question

3 3 A solution containing aspartic acid and lysine is made alkaline.



Aspartic acid



Lysine

This mixture is analysed by thin-layer chromatography.

The moving phase is non-polar. The stationary phase is polar.

Which row is correct?

[1 mark]

	Greater R_f value	**Greater solubility in moving phase**
A	Aspartic acid	Aspartic acid
B	Aspartic acid	Lysine
C	Lysine	Aspartic acid
D	Lysine	Lysine

END OF QUESTIONS

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ANSWER IN THE SPACES PROVIDED

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